Native closed CM relations

P-192 primary; P-224 / P-256 / P-384 / P-521

2026-11-08 · T-1888 · PR 3674

Checked construction

The native continuation now covers all five curves. It passes 150 point/window cases, both full curve-map identities, signed kernel Frobenius actions, the global dual-composition identity, CM norm/degree/label checks, and three orbit fixtures per curve. Eight certificate mutations fail at their intended checks, including a y-sign error with unchanged x action.

Curve	ell	rho	ord mu	signed orbit	norm bits	quotient bits
P-192	13	2	5	10	382	378
P-224	11	10	17	34	451	447
P-256	11	8	3	6	516	513
P-384	19	9	7	14	769	765
P-521	7	6	7	14	1044	1041

Explicit point path

E ⟶ phi ⟶ E1 ⟶ eta ⟶ E2 ⟶ sigma ⟶ E ⟶ correction k

sigma(x,y) = (x / ell², y / ell³); sigma eta phi = [ell].

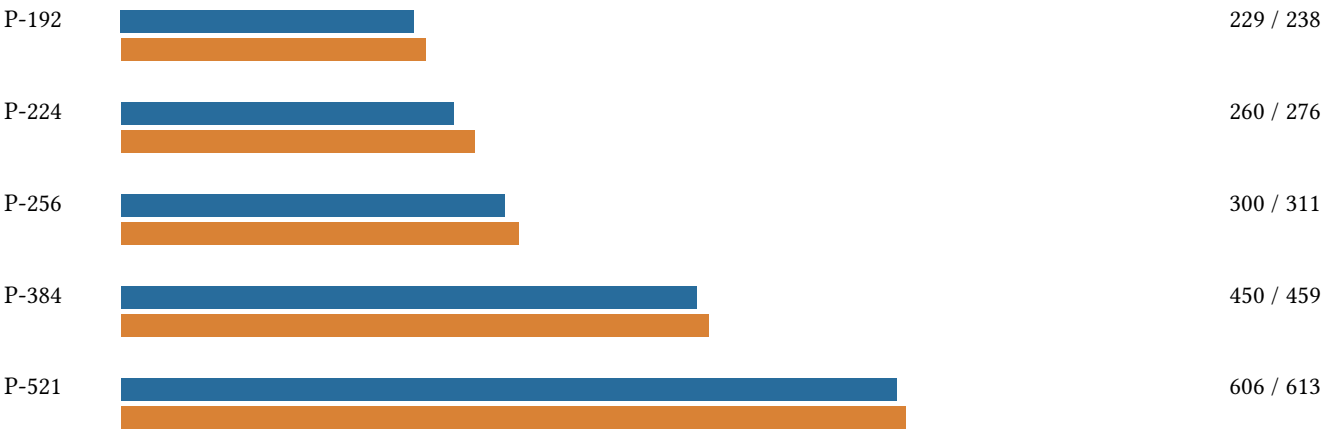
alpha = pi − [c] = a + b pi; a = −c, b = 1; a+b = mu (mod n).

k = mu / ell (mod n); [k] sigma eta phi(P) = [mu]P.

The quotient psi exists globally by kernel containment, with degree norm/ell. Its residual polynomials are unexpanded. The evaluated dual/correction formula is asserted on the rational subgroup and has a different global degree. The native canonicalizer returns the least affine signed-orbit point and its verified scalar transporter.

Exact group schedules: measured event counts, no clocks

Blue: direct public-scalar method. Orange: fused [ell] then [mu/ell]. Each bar is the minimum **unweighted D+A** count over the fixed widths 2–7. This is a source-specific count display, not a runtime optimum.



Data: qualification/structural-v2/rows.jsonl (90 rows, all correct, timing_reported=false). Source/inputs were published before collection; binary and input checks passed. Chart source and exact data are retained together.

Cost vector and accounting boundary

Curve	Direct D,A,N	Fused D,A,N	Map correction D,A,N	Map formulas M,A,I
P-192	190,39,19	195,43,23	191,41,22	114,104,2
P-224	218,42,14	227,49,24	223,47,22	98,88,2
P-256	254,46,23	259,52,27	255,50,25	98,88,2
P-384	380,70,34	387,72,25	383,70,24	162,152,2
P-521	518,88,31	524,89,40	521,88,39	66,56,2

D/A/N count explicit group doubles/additions/negations in the named native routine, including odd-table construction and the fused small-scalar step. The displayed widths are in CONSTRUCTION.md and every scanned width remains in the raw records. The explicit maps add (8ell+10) calls to Field.mul, 8ell calls to Field.add, and two Field.inv calls. This formula follows from the actual full-Horner coefficient-array lengths plus endpoint scales. It excludes backend conversion and setup costs; it is not a complete runtime total.

The full cold timing boundary starts before the required runtime, table and map objects; it includes preparation, maps, endpoint closure, correcting scalar, output verification and destruction. Historical map generation is retained separately. No omitted setup is silently amortized.

Execution and admission record

Check	Observed result
Native qualification	150 point/window cases; 15 orbit fixtures; all five global/signed/label checks pass.
Mutation controls	8 certificate mutations rejected at intended checks; synthetic analysis design rejects 11 additional corruptions.
Local preflight	0.74 and 0.75 other CPU seconds / 2 seconds; limit 0.20. Both refused before cryptographic timing.
Isolation software	Sleep-only completion and SIGTERM restoration pass; always admitted=false.
EKS reconciliation	9 failed jobs, 1 suspended; none restarted or submitted. Direct live read denied at role assumption; live caps and funds unverified.
Timing protocol	Initial two-curve draft passes native schema validation; source/binary freeze and admitted backend remain pending.

The current correction still requires a full-size public scalar. The next algebraic improvement must reduce that work or provide a different directly evaluable closed relation. An admitted end-to-end runtime gain has not been demonstrated; operation counts cannot settle it.

Custody and replay

Source and fixture publication precedes counted data (source commit 7e8109ea2a65; full receipt in structural-v2/publication.txt). New P-192/P-224/P-256 maps were constructed under extension-scope-v4 from published native source 047da426a780. P-384/P-521 preserve historical inputs; the constructor exactly reproduces the saved P-521 map coefficients and scales. P-192 uses its published explicit group because this backend omits the named entry. Earlier compile, backend and projection failures, and the corrected description of the first P-192 failure, remain preserved.

Replay: build native-complete, construct-binary and qualification-tests-binary with the study Makefile; run native –qualify fixtures-v2 and qualification-tests fixtures-v2. Native-v17/v19 contain exact receipts. These are producer-side native exact checks; no independent signed scientific review or Lean formalization is implied.

Standard background: A. Sutherland, MIT 18.783 Lecture 5 (2023), slides 11–16: https://math.mit.edu/classes/18.783/2023/LectureSlides5.pdf . Full derivation, source paths, raw costs and the measurement blocker are linked in CONSTRUCTION.md and problems/cm-marked-scalar-measurement-admission.md.